Bullying and Reading Achievement: A Comparison of South Africa and Brazil Using PIRLS 2021

Chane Watt
Stellenbosch University

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1 Abstract

What is the association between bullying exposure and reading achievement among Grade 4 students in South Africa, and how does this association compare with Brazil? Possible hypotheses: - Students experiencing more frequent bullying have lower reading achievement. - The association remains after accounting for demographic and socioeconomic differences. - The strength of the association may differ by gender or socioeconomic background. - The association may differ between South Africa and Brazil.

2 Introduction

The most prominent headline within the South African education sphere is that 81% of grade 4 learners can not read for meaning. Many studies have attempted to find the underlying mechanisms that resulted in these low test scores with the natural inclination coming to mind being learners social economic status, the school quantile or the confidence of the learners. In 2016 however Anton-Erxleben et al. (2016) found that bullying is one of the key drivers of lower academic performance illustrated in their Directed Acyclic Graph results. This paper will be exploring whether bullying is associated with poorer reading achievement. this will then look at whether some learners are affected more than others, specifically by gender, school quintiles, different Social Economics Status.

On a global level SOuth Africa is performing the lowest of the 57 countries that participated in the PIRLS assignment.

Is bullying associated with poorer reading achievement among South African learners, does this association differ by learner characteristics, and how does the South African pattern compare with Brazil?

Is bullying associated with lower reading achievement in South Africa? Does the association differ by gender, socioeconomic status, or age/over-age status? Is the magnitude and pattern of the

association similar in Brazil?

3 Literature Review

A substantial body of international literature associates exposure to childhood violence with poorer educational outcomes. The systematic review and meta-analysis conducted by Fry et al. (2018) examined evidence from 21 countries and found that childhood violence was associated with lower academic achievement, greater absenteeism and an increased likelihood of school dropout. The authors consequently argue that childhood violence may reinforce educational inequalities among both boys and girls.

More specific evidence on peer bullying is provided by Ozyildirim and Karadağ (2024), who conducted a meta-analysis using TIMSS and PIRLS results from 77 countries. Although the estimated associations were relatively small, bullying had its largest association with reading achievement when compared with mathematics and science. The association was also significantly larger among Grade 4 learners than among Grade 8 learners. These findings make the PIRLS Grade 4 sample particularly appropriate for examining the relationship between bullying frequency and reading achievement. The present study therefore investigates whether this international pattern is also evident in South Africa and Brazil, two upper-middle-income countries with substantially different educational contexts. Whereas Fry et al. (2018) and Ozyildirim and Karadağ (2024) synthesise international evidence from 21 and 77 countries, respectively, this essay focuses on the national patterns observed in South Africa and Brazil. Bullying is particularly prevalent in South African schools: approximately 73–74% of learners in PIRLS 2021 reported experiencing bullying about monthly or about weekly (Mullis et al., 2023). Historical evidence also indicates that bullying is associated with poorer educational achievement. Using PIRLS, prePIRLS and TIMSS 2011 data from Botswana, Ghana and South Africa, Anton-Erxleben et al. (2016) found substantial achievement differences between bullied and non-bullied learners after matching them on observed background characteristics.

Exposure to bullying is nevertheless not distributed evenly among South African learners. Using nationally representative TIMSS 2019 data, Juan et al. (2025) found that age, gender and socioeconomic status were associated with the likelihood of experiencing different forms of victimisation. Boys and older learners generally faced a greater risk, whereas higher socioeconomic status provided some protection against most forms of bullying. These findings support accounting for demographic and socioeconomic differences when estimating the association between bullying and reading achievement.

Brazil provides a useful comparison with South Africa because the two countries share several economic and educational characteristics. Approximately 42% of Brazilian learners reported experiencing bullying monthly or weekly in PIRLS 2021, considerably below the corresponding proportion in South Africa (**?**). Regardless, both countries are upper-middle-income members of BRICS, that have highly unequal income distributions and have implemented comparable education policies (De Villiers and De Villiers, 2023; Spaull, 2023). Nevertheless, Brazil performs approximately

three years ahead of South Africa in reading achievement. This contrast provides a basis for examining whether the association between bullying and reading performance differs between two unequal emerging economies with comparable policy contexts. Evidence from Brazil also supports the approach used in this essay. Using PISA 2015 data, Sbroglio Rizzotto and Aniceto França (2021) found that the relationship between bullying and school performance differed according to both the frequency and type of bullying experienced. Physical bullying and being threatened were associated with lower reading performance, while other forms showed weaker, insignificant or different results. These findings support the use of frequency categories in this essay, as learners who experience bullying weekly may be affected differently from those who experience it monthly or hardly ever. The differences between types of bullying are also important because the measure used in this essay combines several bullying experiences. Therefore, the results cannot show which specific type of bullying has the strongest relationship with reading performance, as discussed in the limitations section. Although Brazil is a useful comparison with South Africa, the findings from these two countries cannot be applied to all emerging economies.

Although Ireland is a substantially different context from South Africa, the longitudinal cohort study by Murphy et al. (2022) provides useful evidence on the relationship between bullying and achievement. Unlike the cross-sectional analysis used in this essay, the study follows the same group of learners over time. The study measures bullying using a binary question at age nine, psychological difficulties at age 13, and achievement at age 15. The findings suggest that bullying at age nine predicted later psychological difficulties, which were associated with lower literacy and numeracy scores. This supports the proposed mechanism in which bullying affects achievement partly through learners' psychological difficulties. Psychological difficulties are therefore not included as a control in the main model, as doing so could remove part of the relationship this essay aims to estimate.

Lastly, this relates to the mechanisms through which Fry et al. (2018) argues that bullying may affect educational achievement. These include absenteeism, reduced school engagement and feeling unsafe at school. These variables are not included as controls because they may form part of the pathway through which bullying is associated with reading scores. Overall, most of the existing literature remains observational and may therefore be affected by reverse causality and differences between learners that are not observed in the data.

4 Data and Methodology

4.1 Data Source and Country Samples

This essay uses data from the 2021 Progress in International Reading Literacy Study (PIRLS), which assesses the reading achievement of Grade 4 learners. PIRLS is conducted every five years by the International Association for the Evaluation of Educational Achievement (IEA) and is widely recognised as an international benchmark of reading achievement and development. The first PIRLS cycle was conducted in 2001, meaning that PIRLS 2021 provides twenty years of reading-trend data for learners at a pivotal stage of schooling, when they are expected to transition

from learning to read towards reading to learn (Mullis et al., 2023). This study compares Brazil and South Africa as two upper-middle-income countries with similar levels of GDP per capita (Spaull, 2023). The dataset is publicly available from the IEA website. The statistical analysis was conducted in `R` using reproducible scripts and the `EdSurvey` package.

4.2 Variables and Measurement

The outcome variable for this study is reading achievement, measured using all five plausible values (`ASRREA01`–`ASRREA05`). Plausible values are multiple draws from each learner's estimated achievement distribution and are intended to produce unbiased population-level estimates rather than precise individual scores. Estimates are calculated across all five plausible values and combined to incorporate uncertainty in the measurement of reading achievement. The main explanatory variable is the bullying-frequency category (`ASDGSB`), which classifies learners as experiencing bullying "About Weekly," "About Monthly," or "Never or Almost Never." The continuous bullying scale (`ASBGSB`) is used as an alternative specification of the main explanatory variable. Higher scores indicate less frequent bullying.

Table 1 reports the variables used in the final analysis. Variables considered during the preliminary audit but not included in the reported models are omitted from the table.

4.3 Empirical Strategy and Specifications

Five models were estimated using progressively larger sets of controls. Model 1 (**M1**) estimates the unadjusted association between bullying frequency and reading achievement. Model 2 (**M2**) adds controls for gender and age. Model 3 (**M3**) additionally controls for home socioeconomic status (SES), while Model 4 (**M4**) adds home language. Finally, Model 5 (**M5**) includes all the preceding variables as well as school location and school disadvantage.

The control variables were introduced in theoretically motivated blocks. Gender was included because previous research indicates that bullying exposure differs between boys and girls (Juan et al., 2025; Anton-Erxleben et al., 2016; Sbroglio Rizzotto and Aniceto França, 2021). Age was included because older learners and learners who have repeated a grade may differ in both their risk of victimisation and academic achievement (Juan et al., 2025). Home resources were added as a measure of socioeconomic background, which is strongly associated with educational achievement and may also be related to bullying exposure (Sbroglio Rizzotto and Aniceto França, 2021). Home language was included because alignment between the language spoken at home, the language of instruction and the assessment language may influence reading performance and may also capture relevant cultural or minority-group differences. Finally, school location and school disadvantage were included to account for differences in school resources, socioeconomic composition and safety environments that may be associated with both bullying and achievement (Anton-Erxleben et al., 2016; Juan et al., 2025).

Because the complete-case requirements become more restrictive as controls are added, the analytical sample decreases in Models 3 and 5. As a robustness check, Models 1–4 were therefore

Table 1: PIRLS 2021 variables used in the analysis

Concept	**PIRLS variable(s)**	**Measurement or use**
Student identifier	IDSTUD	Unique student identifier used for validation and linkage
School identifier	IDSCHOOL	Identifies students attending the same sampled school
Class identifier	IDCLASS	Identifies students attending the same sampled class
Reading achievement	ASRREA01–ASRREA05	Five plausible values for overall reading achievement
Bullying frequency	ASDGSB	Never or almost never, about monthly, or about weekly; main explanatory variable
Student bullying scale	ASBGSB	Continuous scale used for robustness; higher values indicate less frequent bullying
Gender	ASBG01	Student-reported gender; girls are the reference category
Age	ASDAGE	Student age in years
Home socioeconomic status	ASBHSES	Continuous PIRLS home-SES scale; higher values indicate higher socioeconomic status
Home language	ASBG03	Frequency with which the language of the assessment is spoken at home
School location	ACBG05B	Urban, suburban, town, village, or remote-rural location
School economic composition	ACBG03A	Principal-reported percentage of economically disadvantaged students
Student sampling weight	TOTWGT	Produces population-representative estimates
Replication variables	JKZONE, JKREP	Define the jackknife repeated-replication variance design

re-estimated using the complete-case sample from Model 5. Holding the sample constant helps distinguish changes in the estimated bullying coefficients resulting from covariant adjustment from changes caused by differences in sample composition. This common-sample analysis does not eliminate possible selection bias arising from systematic missingness, but it makes comparisons across specifications more informative.

The models are estimated separately for South Africa and Brazil. For country c, plausible value p, and specification m, the estimating equation is

$$Y_{isc}^{(p)} = \alpha_c^{(m)} + \beta_{1c}^{(m)}\text{Monthly}_{isc} + \beta_{2c}^{(m)}\text{Weekly}_{isc} + \mathbf{X}'_{isc,m}\boldsymbol{\gamma}_c^{(m)} + \varepsilon_{isc}^{(p)}, \qquad p = 1, \ldots, 5, \qquad (1)$$

In Equation 1, $Y_{isc}^{(p)}$ denotes reading plausible value p for learner i attending school s in country c. The superscript m identifies the model specification, with $m = 1, \ldots, 5$. The indicators Monthly$_{isc}$ and Weekly$_{isc}$ identify learners classified as experiencing bullying about monthly or about weekly, respectively, with learners who were never or almost never bullied forming the omitted reference group. The corresponding coefficients, $\beta_{1c}^{(m)}$ and $\beta_{2c}^{(m)}$, therefore measure conditional mean differences in PIRLS reading points relative to this reference group. The vector $\mathbf{X}_{isc,m}$ contains the controls included in specification m, while $\boldsymbol{\gamma}_c^{(m)}$ contains their associated coefficients. Finally, $\varepsilon_{isc}^{(p)}$ is the error term. The models are estimated separately for South Africa and Brazil and across all five plausible values.Two additional specifications interact bullying frequency with gender (I1) and home socioeconomic status (I2) to assess whether the bullying–achievement association differs across these groups.

4.4 Survey Design and Estimation

All estimates incorporate the PIRLS total student sampling weight (TOTWGT), which accounts for unequal probabilities of selection and adjustments for nonresponse at the school, class and student levels. Sampling variances are estimated using the PIRLS jackknife repeated replication procedure and the supplied JKZONE and JKREP variables. This procedure accounts for the clustered and stratified survey design. The released data produce 250 replicate estimates for South Africa and 196 for Brazil. Reading-achievement estimates additionally incorporate uncertainty across the five plausible values. The analysis was implemented in R using the EdSurvey package.

4.5 Methodological Limitations

Several limitations should be considered. First, bullying exposure and reading achievement were measured during the same assessment period. The analysis therefore cannot determine which occurred first. Bullying may contribute to poorer achievement, but lower-achieving learners may also face a greater risk of victimisation. Unobserved family, peer and school characteristics may additionally influence both bullying and achievement (Anton-Erxleben et al., 2016).

Second, bullying exposure is self-reported and may therefore be affected by recall error, differences in learners' interpretation of the questions and their willingness to report victimisation.

Third, the common-sample analysis reduces the number of observations because all models are re-estimated using only learners with complete information for Model 5. This allows changes in the bullying coefficients to be compared without simultaneously changing the composition of the sample. However, excluding learners with missing information may introduce selection bias if missingness is related to bullying or reading achievement. The smaller sample may also result in less precise estimates.

Fourth, South African and Brazilian Grade 4 learners were assessed one year later than originally scheduled, during a period of widespread COVID-19-related educational disruption.

Finally, Brazil and South Africa should not be treated as interchangeable simply because both are classified as upper-middle-income countries. They differ in their education systems, levels and patterns of inequality, linguistic environments, school conditions and social understandings of bullying. The estimated associations are therefore context-specific averages and may conceal substantial heterogeneity across learners and schools. Evidence from these two countries cannot be generalised to all upper-middle-income countries.

5 Results

Descriptively, approximately 73–74% of South African learners were classified as experiencing bullying about monthly or weekly, compared with approximately 42% of Brazilian learners. The regression analysis below examines whether the association between bullying and reading achievement persists after progressively introducing student, household, and school-level controls.

Table 2 presents the M1–M5 estimates for South Africa and Brazil. There is a negative relationship for both Brazil and South Africa between reading scores and bullying penalties. South Africa in Model 1 that just controls for Bullying Penalty indicates a decrease of 55.1 points in reading scores if the learner is bullied on a monthly basis and a decrease of 116.6 points if bullied on a weekly basis. Brazil displays the same negative pattern: learners bullied monthly score 36.4 points lower, while those bullied weekly score 123.5 points lower. All four M1 bullying coefficients are statistically significant at the 1% level.

The model develops in interesting directions as controls are added in M2, M3, M4 and M5. In South Africa, the estimated bullying penalties become progressively smaller as controls are added, with the M5 coefficients indicating a 36.1-point decrease for monthly bullying and an 80.6-point decrease for weekly bullying. Brazil, by contrast, remains comparatively stable: the monthly penalty increases in magnitude from 36.4 points in M1 to 38.4 points in M5, while the weekly penalty decreases from 123.5 to 113.3 points. The standard errors generally decrease as controls are introduced, indicating greater precision. The explained variance increases from 13.4% to 38.0% in South Africa and from 13.9% to 35.5% in Brazil.

Table 2: Bullying and reading achievement: main model sequence

	South Africa					Brazil				
	M1	M2	M3	M4	M5	M1	M2	M3	M4	M5
Bullied about monthly	-55.1***	-53.8***	-46.8***	-43.9***	-36.1***	-36.4***	-37.5***	-39.3***	-36.5***	-38.4***
	(6.9)	(6.8)	(5.9)	(5.5)	(5.5)	(5.7)	(5.4)	(5.5)	(5.3)	(5.8)
Bullied about weekly	-116.5***	-110.4***	-97.9***	-94.2***	-80.6***	-123.5***	-119.2***	-112.9***	-109.4***	-113.3***
	(7.8)	(7.5)	(6.8)	(6.7)	(6.7)	(7.6)	(8.0)	(7.3)	(7.2)	(7.9)
Boy		-44.4***	-43.4***	-40.8***	-43.1***		-10.2**	-5.4	-3.9	-4.9
		(3.1)	(3.5)	(3.8)	(3.9)		(5.0)	(4.6)	(4.5)	(4.8)
Age (years)		-10.0***	-7.6**	-6.9**	-7.6**		-24.6***	-13.1***	-12.6***	-12.7**
		(3.0)	(3.4)	(3.4)	(3.3)		(3.9)	(4.4)	(4.6)	(4.8)
Home SES (ASBHSES scale points)			25.7***	24.9***	16.8***			26.3***	25.7***	21.6***
			(2.3)	(2.4)	(2.5)			(2.0)	(2.0)	(1.8)
Home language: almost always test language				7.4	-5.5				-18.7*	-20.8*
				(8.1)	(8.1)				(10.7)	(10.7)
Home language: sometimes test language				46.3***	22.4***				7.6	12.3
				(7.4)	(7.1)				(9.8)	(9.5)
Home language: never test language				-28.0***	-40.7***				-93.2***	-83.3***
				(8.9)	(10.6)				(19.2)	(19.4)
School location: suburban					-18.0					-1.3
					(17.6)					(15.8)
School location: medium city/large town					1.9					-18.6**
					(34.0)					(9.1)
School location: small town/village					-45.1**					-25.3**
					(18.8)					(11.5)
School location: remote rural					-55.5***					-14.4
					(16.5)					(25.3)
Economically disadvantaged: 11 to 25%					-49.4*					-25.5**
					(27.8)					(10.9)
Economically disadvantaged: 26 to 50%					-82.8***					-47.6***
					(20.2)					(13.7)
Economically disadvantaged: more than 50%					-117.3***					-40.9***
					(17.1)					(11.7)
Other gender category							-100.2***	-86.7**	-85.6**	-49.1**
							(31.1)	(35.1)	(35.6)	(23.6)
Learners	11,660	11,618	9,167	8,505	7,177	4,715	4,706	4,110	4,022	3,563
R^2	0.134	0.170	0.278	0.296	0.380	0.139	0.165	0.317	0.324	0.355

Notes: Entries are coefficients with jackknife standard errors in parentheses. The reference group is almost never bullied; the gender reference is girls. All models use the default PIRLS student weight and all five reading plausible values. M1 includes bullying; M2 adds gender and age; M3 adds continuous home SES; M4 adds home language; M5 adds school location and economically disadvantaged composition. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Figure 1 summarizes the bullying coefficients across M1–M5, showing how the estimated associations change as successive controls are introduced.

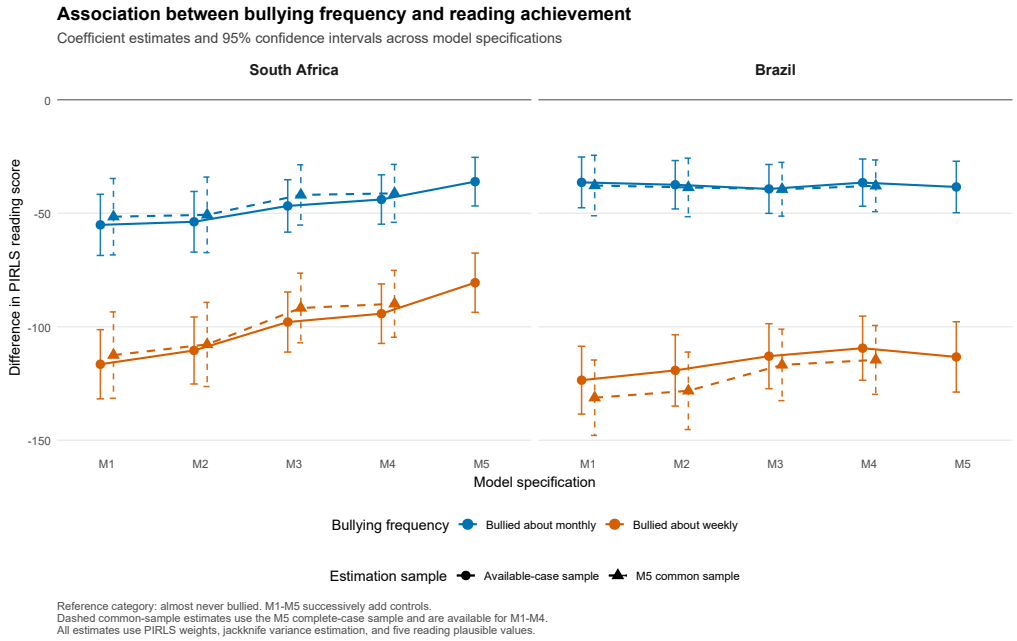


Figure 1: Estimated association between bullying frequency and reading achievement across model specifications

Re-estimating M1–M4 on the common M5 sample produced a similar pattern of coefficients, suggesting that the changes across models were not primarily attributable to sample composition (Appendix Table 4). The continuous-scale specification also produced results consistent with the main analysis (Appendix Table 5).

Table 3 presents the interactions between bullying and gender (I1) and between bullying and home SES (I2). The gender interactions are not statistically significant in either South Africa or Brazil. There is therefore no evidence that the association between bullying and reading achievement differs between boys and girls.

The SES interaction produces a different pattern. In South Africa, each one-unit increase in home SES makes the estimated monthly bullying penalty 13.2 points more negative and the weekly penalty 20.3 points more negative. In Brazil, the corresponding interaction coefficients are positive: 3.2 points for monthly bullying and 9.0 points for weekly bullying. The monthly interaction is not statistically significant, whereas the weekly interaction is significant at the 5% level. Thus, there is evidence that higher SES intensifies the bullying penalty in South Africa, while the Brazilian estimates provide limited evidence that higher SES moderates the weekly bullying penalty.

Table 3: Heterogeneity in the association between bullying and reading achievement

	South Africa		Brazil	
	I1	I2	I1	I2
Bullied about monthly	-46.5***	-42.2***	-34.1***	-39.8***
	(7.0)	(5.7)	(8.8)	(5.6)
Bullied about weekly	-95.3***	-95.5***	-120.4***	-110.9***
	(7.8)	(6.4)	(10.6)	(6.9)
Boy	-41.2***	-42.9***	-4.1	-5.6
	(6.7)	(3.4)	(4.9)	(4.7)
Age (years)	-7.6**	-8.0**	-12.9***	-13.5***
	(3.4)	(3.4)	(4.4)	(4.3)
Home SES (ASBHSES scale points)	25.7***		26.2***	
	(2.3)		(2.1)	
Bullied about monthly × Boy	-0.7		-10.5	
	(9.4)		(11.1)	
Bullied about weekly × Boy	-5.1		9.9	
	(7.4)		(12.7)	
Home SES (centred scale points)		37.5***		24.1***
		(2.8)		(2.0)
Bullied about monthly × Home SES (centred scale points)		-13.2***		3.2
		(2.8)		(3.0)
Bullied about weekly × Home SES (centred scale points)		-20.3***		9.0**
		(2.8)		(4.5)
Other gender category				-86.7**
				(35.2)
Learners	9,167	9,167	4,076	4,110
R^2	0.278	0.289	0.314	0.319

Notes: Entries are coefficients with jackknife standard errors in parentheses. I1 interacts bullying with gender and excludes Brazil's 41 learners in the Other gender category. I2 interacts bullying with continuous ASBHSES centred at the design-weighted country mean; one unit is one PIRLS ASBHSES scale point. The reference bullying category is almost never and the gender reference is girls. Significance: *** p<0.01, ** p<0.05, * p<0.10.

6 Conclusion

A Sensitivity and robustness analyses

Table 4: Common-sample sensitivity analysis

	South Africa				Brazil			
	M1	M2	M3	M4	M1	M2	M3	M4
Bullied about monthly	-51.5***	-50.7***	-41.9***	-41.2***	-37.8***	-38.6***	-39.4***	-37.9***
	(8.6)	(8.5)	(6.8)	(6.5)	(6.8)	(6.6)	(6.0)	(5.8)
Bullied about weekly	-112.5***	-107.8***	-91.7***	-89.9***	-131.2***	-128.2***	-116.8***	-114.6***
	(9.7)	(9.5)	(7.8)	(7.5)	(8.5)	(8.7)	(8.0)	(7.7)
Boy		-45.5***	-43.1***	-41.4***		-5.9	-4.5	-3.2
		(4.3)	(4.1)	(4.1)		(5.7)	(4.9)	(4.8)
Age (years)		-5.9	-5.5	-6.6*		-24.7***	-13.8***	-13.5***
		(3.9)	(3.7)	(3.6)		(5.2)	(4.7)	(4.7)
Home SES (ASBHSES scale points)			26.5***	25.4***			25.7***	25.7***
			(2.6)	(2.6)			(2.2)	(2.2)
Home language: almost always test language				9.1				-20.2*
				(9.2)				(11.4)
Home language: sometimes test language				46.8***				12.4
				(7.9)				(9.6)
Home language: never test language				-24.6**				-87.7***
				(10.1)				(19.4)
Other gender category						-72.9***	-53.4**	-50.2**
						(24.3)	(24.0)	(24.4)
Learners	7,177	7,177	7,177	7,177	3,563	3,563	3,563	3,563
R^2	0.123	0.156	0.275	0.296	0.156	0.177	0.317	0.328

Notes: Entries are coefficients with jackknife standard errors in parentheses. All models use the M5 complete-case sample. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 5: Continuous-bullying robustness model

	South Africa	Brazil
	R1	R1
Bullying scale (higher = less bullying)	18.7***	16.8***
	(1.3)	(1.2)
Boy	-43.7***	-5.0
	(3.6)	(4.8)
Age (years)	-8.0**	-13.7***
	(3.4)	(4.9)
Home SES (ASBHSES scale points)	26.1***	27.4***
	(2.3)	(2.1)
Other gender category		-97.7**
		(36.8)
Learners	9,167	4,110
R^2	0.270	0.293

Notes: Entries are coefficients with jackknife standard errors in parentheses. Higher ASBGSB values indicate less bullying. Significance: *** p<0.01, ** p<0.05, * p<0.10.

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